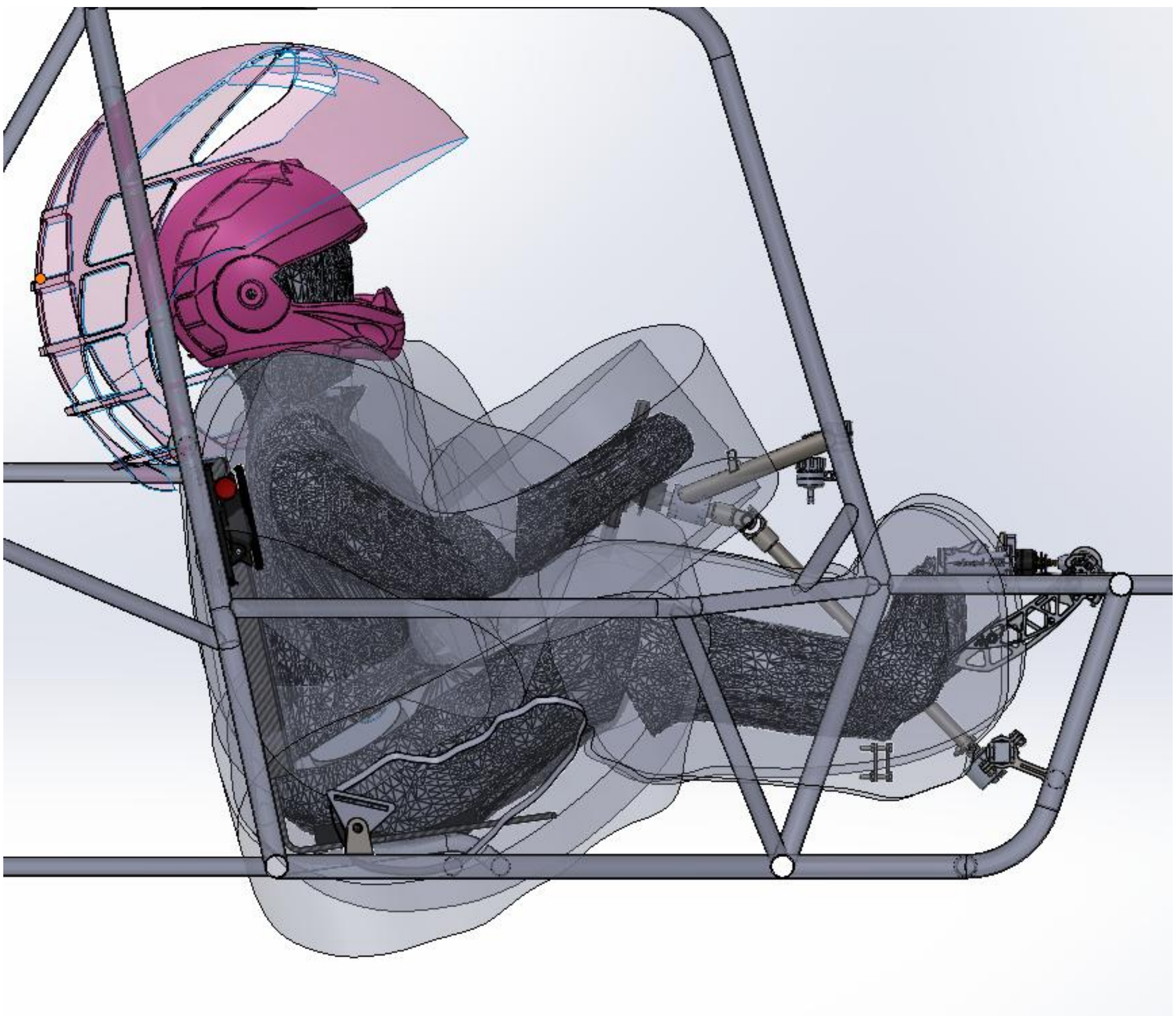
Clearance Models

LiDAR-to-Model Workflow

Version 1.0 - July 19, 2026



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Section 1 Project Overview

1.1 Background

Driver packaging is essential to creating a competitive and rules-legal vehicle. Optimization in this area allows for minimum vehicle dimension, contributing to a lightweight vehicle.

Technical inspection for the competition, additionally, must be completed and passed in order to compete. In M24 (May 2025), only one driver was suitable for competition, as the rules state that laterally, "The driver's helmet shall have 152 mm (6 in.) clearance, while the driver's shoulders, torso, hips, thighs, knees, arms, elbows, and hands shall have 76 mm (3 in.) clearance." Vertically, "The driver's helmet shall have 152 mm (6 in.) minimum clearance from any two points among those members that make up to top of the roll cage" (Baja SAE Rules 2026).

M24 attempted to compete with multiple drivers for endurance (3hr race), and only one 5'5" driver was able to be accommodated, while others failed technical inspection.

Previous methods of driver packaging used scalable general models called "Wilsons," imaged in Figure 1. The issue is each person has different proportions, so a 5" model may not match up well with a 5" driver, leading to driver disqualification.

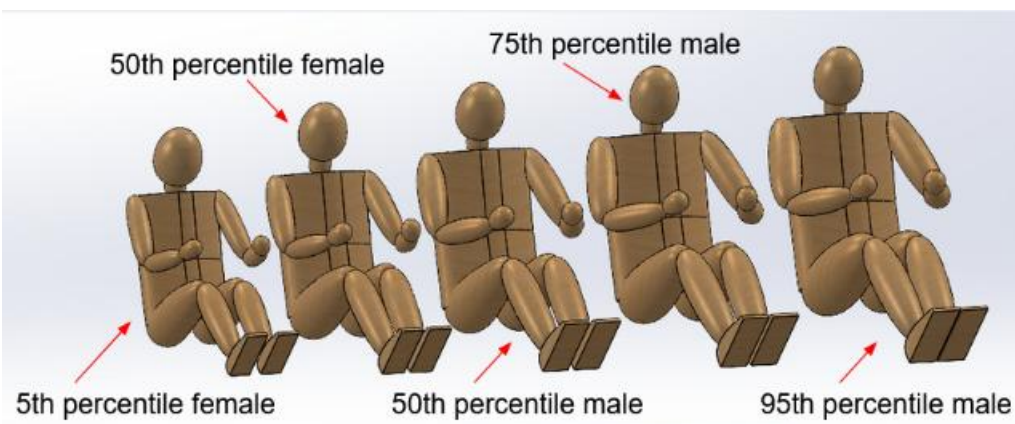


Figure 1: Wilson models used to determine driver clearance. All are same proportion and scaled to match the height of a potential driver.

1.2 Project Objective

I was tasked with creating accurate, personalized driver models for each potential driver. The models were also desired to be moveable, which would allow for a greater range of positions. This range could create multiple potential driver configurations, allowing for better optimization capacity and potentially more drivers. For ease of visualization, clearance shells, which could visually represent the necessary clearance for technical inspection, were desired to allow for quick configuration optimization and iteration.

1.3 My Role

I was required to make the objective a reality. There was no guidance given except for the desired outcome and the LiDAR scans of the drivers, as I was not available in-person.

1.4 Final Outcome

After several weeks of iteration and research, I developed a LiDAR to Moveable Model workflow, which allowed for the quick transformation of scan to model. The models I created were used to package drivers and were validated by an ergonomic jig I helped design and fabricate. M25 technical inspection for driver clearances was passed with ease with all 5 drivers.

Section 2 Problem Definition

2.1 Existing Driver-Packaging Methods

As explained earlier, the driver packaging method used previously was by using static Wilsons (Fig. 1). Other approaches could be taken, such as using specific software to model anthropomorphic ergonomics. RAMSIS is a widely used software used in the commercial automotive and aerospace industries, which is immensely powerful, and it was the software that got me interested in the Project Engineer position for ergonomics.

Unfortunately, we couldn’t secure sponsorship that would allow us to use the software, which made it so that a modeling method would have to be created using software already on hand.

2.2 Project Scope

The project included the following:

- Scans of potential drivers
- Measurements of driver limbs
- Scaling the LiDAR scans
- Simplifying the mesh data
- Segmenting the limbs
- Creating appropriate joint locations
- Building an articulated CAD model
- Adding clearance geometry
- Subteam discussion and compromise for integration

Section 3 Workflow Overview

3.1 Summarized Process

First, LiDAR scans of each driver were taken in a driving position (Figure 2). The scans are then imported and scaled in SolidWorks. After verification of geometry and proportions, the scans are simplified for quicker processing time, with insignificant geometrical consequences. Each limb is then sectioned and saved as a part file, with precise cuts at joint areas. The parts are then assembled in SolidWorks using a ball and socket joint. Now that the movable model is completed, each limb is then bisected by a reference plane. The intersection of the part and the plane yield an outline, which is then offset by 3 inches. The sketch is then extruded in both directions by 3 inches past the part. To meet the vertical and lateral requirements of the head, a helmet was fixed in position to the head. A copied helmet was added and scaled up to a 6-inch normal displacement from the fixed helmet, as a simple scaling of a spherical shape works as intended, while a scaling of a non-spherical object introduces unwanted displacement (see Design Iterations for more details).

3.2 Software

- Polycam
  - Used for LiDAR scans
  - Has a 1-week free trial for each device
  - I was sent scans taken from this software and also experimented personally with the software
    - I had my family scan me as a potential driver, as I am within the height range (5’7”).
- SolidWorks
  - Used for scaling, assembly, and clearance shell creation
  - Was the required “result software,” as the car was entirely modeled in SolidWorks
- MeshLab
  - Used for quick geometric simplification to reduce computational expense in SolidWorks
  - Was required as importing a mesh to SolidWorks occasionally yielded an error of “Timestamp of file is invalid,” where geometric simplification solved the error
- Fusion 360
  - SolidWorks does not support simple mesh manipulation. At the very least, it does not support it as well as Fusion 360
  - Fusion 360 is free, and there is a mesh tab, of where a plane can be created, the model can be positioned with ease and accuracy, and the intersection between the plane and the model can split the model in two
    - Repeat this process till all body parts are separated

3.3 LiDAR Background

LiDAR stands for Light Detection and Ranging. It sends rapid laser pulses at a target and calculates the time required for the reflected light to return. LiDAR cameras are built into newer iPhone and iPad models, which allowed for ease of use for us students. Highly reflective or transparent surfaces disrupt the sensor.

The precision of the scanners we used is within 1cm, which is sufficient for our uses. The precision largely depends on number of scan revolutions and distance between scanner and object.

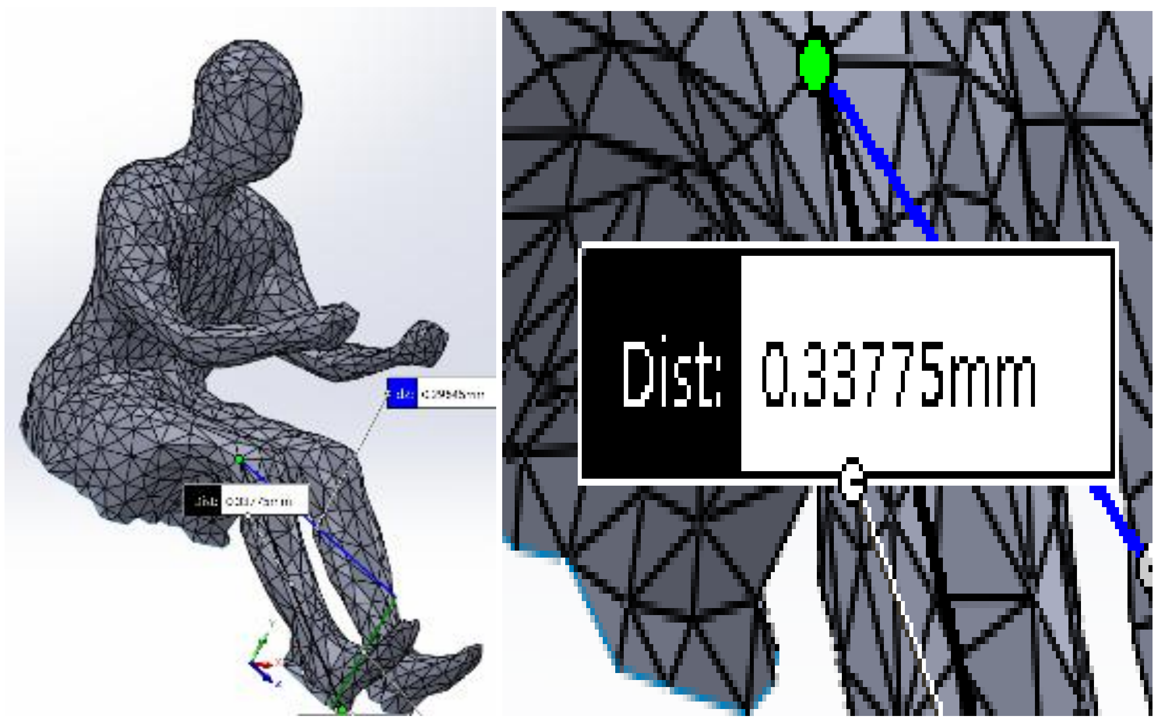
3.4 Taking LiDAR Scans

First, the potential drivers assumed a driving position (at this point the idea for movable model was not proposed). The person tasked with scanning revolves several times around the subject, who must remain very still, and scans very close, ensuring that all areas of the subject have been adequately scanned.

Figure 2: LiDAR scan of me in summer of 2025. Background cropped, such that there are small holes at base of feet and butt. *Shaved head not necessary, but did help with actual head shape.

3.5 Manipulating LiDAR Scans

The scan was then imported into SolidWorks, of which an error presented itself as "Timestamp of file is invalid." After simplification in MeshLab (see 3.6), the model was successfully imported into SolidWorks, where the scaling was completely inaccurate. Measurements of the real drivers' limbs were required to accurately scale up the model. To test the accuracy, I scaled up a model taken of me in driving position based on limb measurements and compared it to a second model of me with a ruler between my hands in driving position. The difference was insignificant, but there was a slight offset, leading to my proposal in the Future Improvements section. The reason my proposal was not utilized in M24 was due to the expiration of the Polycam trial, and the relatively insignificant nature of the uncertainty.



(a) (b)

Figure 3: LiDAR scan as mesh in SolidWorks. (a) This is the model, to be scaled, which was taken from Fig 2. (b) This is a zoom of (a), detailing the magnitude of the scaling factor required to get a usable model.

3.5 Initial Demonstration

After scaling the models I was given, I overlayed the models in order to demonstrate a theoretical drivers range. I made it so the farthese part of the models back was mated to a plane, and mated the lowest part of the butt to a perpendicular plane. I then later put the models in the previous year's (M24) car.

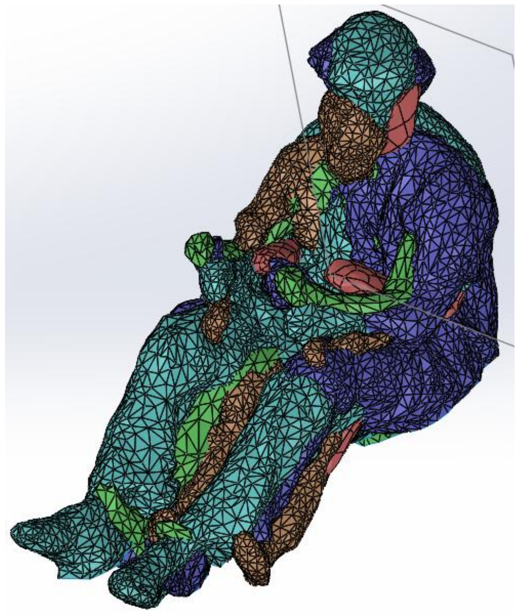


Figure 4: Overlayed, scaled models.

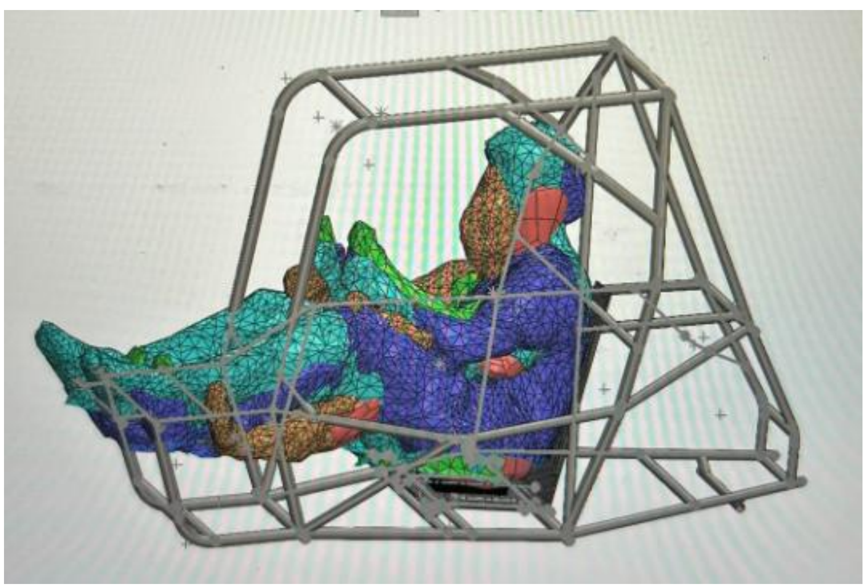


Figure 5: Overlayed models in M24 car, informing of potential drivers and incompatible drivers.

3.6 Mesh Simplification

Simplification of the mesh is incredibly important for SolidWorks speed and functionality. A time when I didn't simplify, SolidWorks took nearly 10 seconds to process and attempt to change the view.

As stated earlier, mesh simplification can also allow for the actual manipulation of the mesh in SolidWorks, as the program may throw seemingly random errors for not having a simplified mesh.

For a detailed guide I wrote after discovering the process:

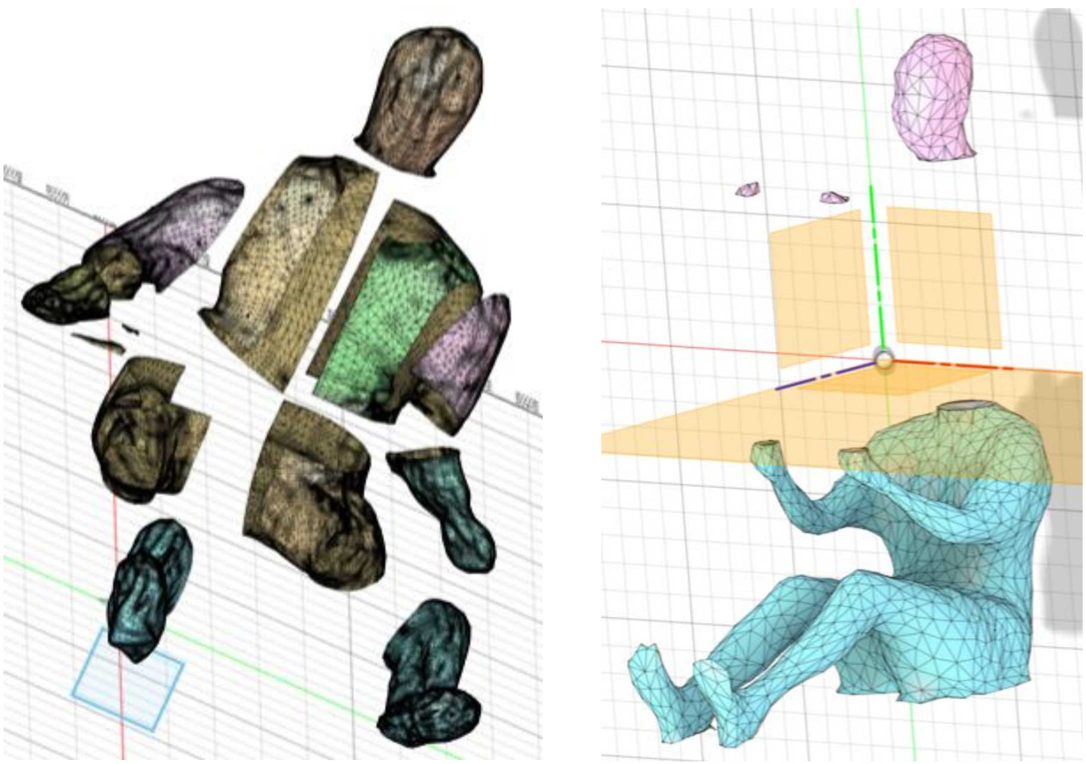
"Go to Filters > Remeshing, Simplification and Reconstruction > Simplification: Quadric Edge Collapse Decimation. For adequate speed, ensure the target number of faces is below 10000. If the number of faces is already below, then any number below the initial number, as long as the mesh maintains its required geometric integrity, is adequate and apply may be pressed. The figure is less detailed but significantly easier to handle.

The next step is in the same place: Filters < Cleaning and Repairing. First, apply remove unreferenced vertices, then remove duplicate faces, and finally merge close vertices. Finally, go back to file, and export mesh as STL. The file should then be able to function within SolidWorks without significant computational expense."

3.7 Body Segmentation

The procedure is simple if the user has experience with CAD software. Within Fusion, there is a tab near the top, which allows the user to go between multiple model types. After choosing mesh, a plane must be created, the mesh must be moved into position, then the MeshPlaneCut functionality must be used. Repeating this procedure for each desired limb and joint is necessary. To separate the legs, I cut the entire body in half, then mated the torso in SolidWorks, despite there being no need to cut the torso in half besides expedience.

Because the mesh is hollow, a simple method of patching the hole generated by the cut is to use the MeshRepair function.



(a) (b)

Figure 6: Segmentation of models (a) Richard and (b) Nathan. (a) Shows the final result of segmentation, yielding the desired parts required for SolidWorks Assembly. (b) Shows the first step of segmentation, where the head intersects the plane at the neck, and MeshRepair is used to patch the holes. Because the plane is infinite, parts of the hand was also cut, which would be deleted in SolidWorks, as the top portion of the hand has no clearance issues.

3.8 Model Assembly

The assembly process is relatively straightforward. I created a simple ball joint, using a rectangular prism and a sphere. The mesh in SolidWorks, after MeshLab, is treated as a solid, so the surface is able to be be cut. I cut a depth in one limb of the extrusion plus the radius of the sphere for the square peg. This allows for surface contact, as the revolved extrusion cut is directly in the main body, which is the already established part, such as the torso.

(a) (b)

Figure 7: Assembly of Richard model. (a) This is a zoomed in picture of the assembly, focusing on the upper body. In the bottom, it is clear that there is separation between the legs, as the legs are in a stretched position. The head is a ball joint attached to an extruded cylinder, as the plane of the mesh was difficult to extrude properly at times. (b) This shows the attachment procedure. The revolved joint is cut out for the ball to be concentric and have surface contact, while the prism will be fully inserted and constrained within the arm.

3.9 Clearance Shells

This section taught me a lot about SolidWorks' toolset. To create a clearance shell, a reference plane must be placed directly center of the object, bisecting it. After, an intersection curve can be applied for ease of modeling. Rather than offset the curve by 3 inches, as sharp edges tend to "runaway," 3-inch, normal contruction lines are placed at different displacements and inclines to allow for a spline to be drawn.

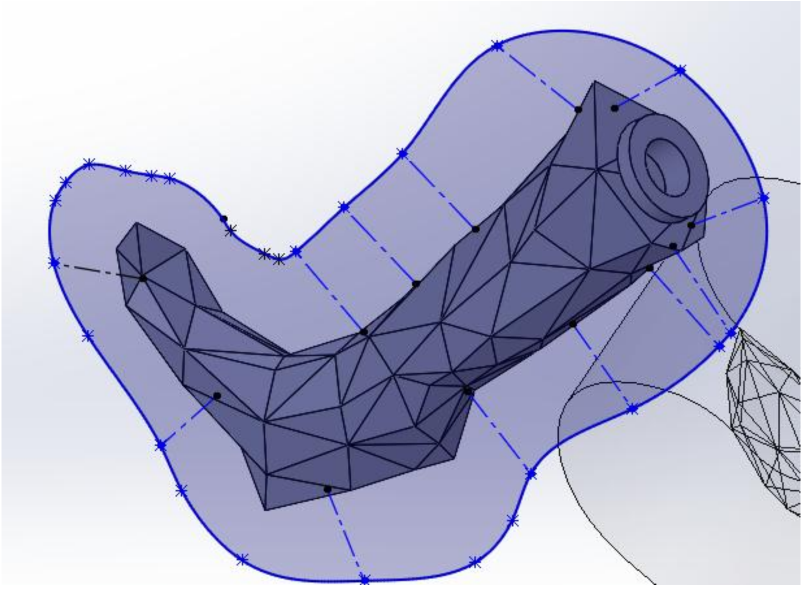


Figure 8: Three-inch clearance spline sketch for vertical clearance, incorporating 3 inch normal lines.

The sketch is then extruded 3 inches past the surface of the limb on both sides. To properly show the clearance, it is beneficial to make the shell transparent.

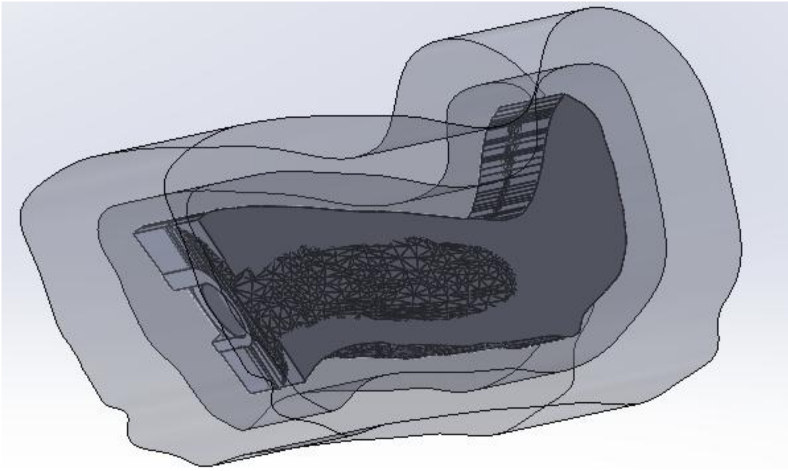


Figure 9: Clearance shell of both 1 inch and 3 inch. This is a proof of concept, as all the other clearance shells are strictly 3 inches.

After applying the 3 inches to every limb, the helmet rule must have 6 inches laterally and veritically. This process is much simpler, as scaling an object is similar to blowing it up. With a part like a leg, if it were to be scaled, it would not only be scaled 3 inches up, but also forward, which would lead to a diagonal offset. The offset for a spherical object, however, is perfectly suitable for our needs. Simple trial and error to determine the scale needed to create a normal distance between identical points on the helmets was utilized.

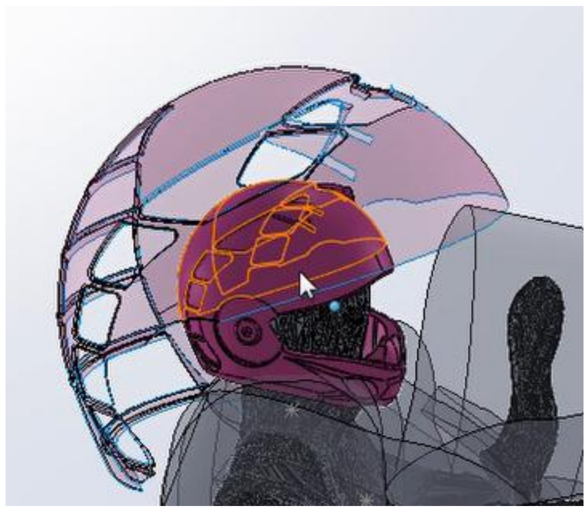


Figure 10: Helmet with clearance helmet shell, allowing for clear lateral and vertical clearance determination.

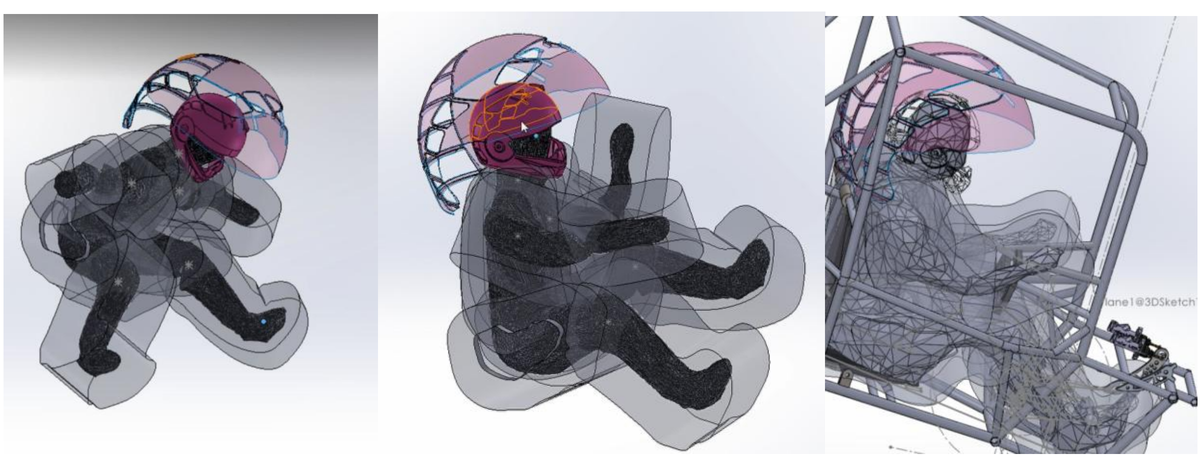


Figure 11: Fully articulated model with clearance shells

Section 5 Collaboration and Integration

Collaboration was fruitful, but difficult. Integrating all of the different subsystems and their respective components proved to be difficult, as many compromises were required. Assisting chassis with their design was simple, as my models set a lower limit for the chassis size. This did, however, require discussions about a lower limit for comfort, as drivers could alter their angels in order to yield superior clearance at the cost of their comfort, allowing for a smaller chassis and weight reduction. Looking through NASA's ergonomic document, I determined a lower limit for chassis size, then validated it, explained in section 6. The upper limit for chassis size is not necessary, as it is significantly greater than anything the team would be willing to create, as a larger car meant a non-competitive car.

Other discussions were with power powertrain and within Driver Interface as to the steering wheel and pedal placement, of which I would later be responsible for determining.

Section 6 Verification and Validation

Validation came in the form of an ergonomic jig, of which documentation is available on my website. In short, the ergonomic models were used to determine chassis geometry, upon which all subsystems relied upon. In CAD, my Co-PE and I developed an ergonomic jig which allowed for simple measurements of the steering wheel and pedal position based on preferred driver angels. I designed the steering wheel locking mechanism, and the pedal's rotation and translation features.

Figure 12: Articulated model in ergonomic jig model, ensuring jig model adequately maps the chassis design, while allowing for ease of adjustment for potential chassis alterations.

Section 7 Results

The models allowed for 4 additional drivers, with an increase of driver capacity from a maximum of 5’5” to 5’9”, while also being more comfortable in comparison to M24. M25 was also a lighter car compared to M24, so the modest size increase did not drastically affect weight, as it was still kept the minimal volume, achieved through the articulated, clearance models.

Section 8 Workflow Iterations

My first attempts were to use only SolidWorks, as it was the software I was most familiar with. However, through a lot of trial and error, requiring me to seek out different methods, I eventually discovered the necessary softwares needed to expedite the process.

4.1 The Loft Intersection

My goal was to create the segmented parts within SolidWorks. However, the mesh was unusable at the time, so I decided to create a reference plane at where the joint would be, and create incremented reference planes, each with an intersection curve. I then lofted the curve to produce the part, which was aesthetic, but it was only practical for simple geometry like the arms.

4.2 Mesh-to-Part

Similar to 4.1, I used loft and intersection curves. However, this time I believed it to be more efficient to loft the entire figure, then to cut it up later. I began my loft from the base of the model, however it proved to be challenging, as the complex geometry of the figure necessarily makes the loft poor quality, as well as the issue with lofting multiple closed bodies on the plane due to the legs only intersecting starting on the 3rd or 4th plane.

4.3 Blender Boolean

Still struggling, I attempted to use Blender and its Boolean function to separate the model’s limbs. I cannot remember specifically why this did not work, but Blender is an immensely challenging software, so I assume it is due to spending a significant amount of time on the software and not making progress. This could potentially work, but because of the high skill floor, I determined it to be impractical.

Section 9 Future Improvements

9.1 Standardize LiDAR Scans

The LiDAR scans I received had potential drivers all assume the same position, however the angle between their limbs and hand placement varied substantially. Additionally, with the lack of a clear metric within the scan, scaling relied on faulty measurements, which increased the uncertainty. For example, I asked for multiple limb measurements with instructions on the measurement process, but scaling based on one metric yielded a fairly substantial difference when compared to the other metric.

With the goal of creating articulated clearance models, a more practical approach is to have the driver hold a position with arms stretched, similar to a wingspan check, and standing with legs very slightly bent. The feet a ruler's distance away, or a known distance apart, while also knowing the wingspan of the driver, will allow for significantly accurate scaling.

9.2 Elaborate and Deliberate Mesh Processing

The approach I took was "what works, works." For our purposes, this approach was great, as the mesh quality was not the bottleneck, and it was more effective to put my time into other matters, rather than this.

However, improving processing speed and being sure the mesh geometry remains exactly as needed would be an improvement that could be worth pursuing as a "nice-to-have."

9.3 Accurate Joint Placement

The positioning of the joints were relatively arbitrary, with my decisions being guided more by trail and error, and "what looks right." However, a rotation point at the top of the shoulder and the bottom of the shoulder yield different potential paths, which alters the accuracy of the model.

Learning the biomechanics of a human and the anatomy of where joints are would aid in the accuracy of placement. For our purposes, the relatively large uncertainty in joint position did not yield significant enough path differences to consider spending more time on this issue.

9.4 Improved Clearance Geometry

The clearance geometry is extremely approximate and conservative, as the entire limb is encased in a clearance shell that is the minimum distance from the farthest part on the model. This means that the clearance of parts of the limb exceed the 3 inch minimum distance, sometimes having clearance of up to 5 inches, which could potentially the weight cuts associated with a potentially smaller chassis.

Improving clearance geometry by ensuring the displacemet from each triangle is 3 inches would allow for a more accurate clearance model, however it is unclear how to achieve this without expensive software. A potential idea is to utilize my method, but also create a sketch on a perpendicular plane, which would be more accurate the a uniform extrusion.

Section 10 References

1. Baja Rules 2026 – Only PDF available (www.bajasae.net)
2. NASA -- Ergonomic Models of Anthropometry, Human Biomechanics, and Operator-Equipment Interfaces (https://ntrs.nasa.gov/api/citations/19890017973/downloads/19890017973.pdf)

Section 20 Document Revision History

Version	Date	Author	Change
1.0	7/19/2026	Nathan Robinson	Document Release